



ANKIT GHOSH

ag9822@srmist.edu.in | 6294230462 | 2027

B.Tech, Artificial Intelligence

Address : mullai boys hostel, pothéri, tamil nadu, chennai

Summary

Results-oriented Machine Learning / NLP Engineer with strong practical expertise in Natural Language Processing and Transformer-based architectures, delivering scalable and production-ready AI systems. Highly proficient in Python, PyTorch, NumPy, Pandas, and scikit-learn, with proven end-to-end ownership across data processing, model training, evaluation, optimization, and deployment. Deep understanding of Transformers, attention mechanisms, text summarization, and embeddings, including hands-on experience fine-tuning and deploying models such as T5. Solid foundation in Data Structures & Algorithms (Python and Java) and system-level design for robust ML pipelines. Actively expanding into Computer Vision and multimodal learning to build comprehensive AI solutions. Well-prepared to contribute immediately to high-impact engineering teams at Microsoft, Amazon, or Google with clean, reliable, and scalable machine learning systems.

Education

SRM IST-Kattankulathur B.Tech · Artificial Intelligence	2027 CGPA - 8.20/10
H.S.Memorial School 10th · Computer Science · Janai, West Bengal	Mar 2020 - Jun 2021 8.89
Swarajya Senior Secondary School 12th · Computer Science · Alwar, Rajasthan	Feb 2022 - Jul 2023 7.78

Experience

PRS Neurosciences https://prsneurosciences.com/ ML Intern · Intern · Full-Time	Jun 2025 - Present Virtual
- Built an automated image preprocessing pipeline with pose-based subject isolation and background reduction to standardize human-centric datasets. - Optimized storage and performance using Guetzli and MozJPEG compression while preserving model-ready image quality. - Designed a modular, scalable workflow supporting single-image and batch processing, improving efficiency for downstream computer vision tasks.	
Sasken Technologies Link Software Developer · Full-Time	Dec 2025 - Present Sector V, Bidhannagar, Kolkata, within the Bengal Intelligent Park
Developed a zero-shot conversational learning framework for a robotic system, enabling real-time knowledge validation and incremental learning from human speech input. Built a multilingual speech understanding and natural language processing pipeline supporting speaker-independent recognition, sentence normalization, and intent extraction. Integrated cloud-based large language model (LLM) inference with on-device memory systems, enabling scalable, context-aware, low-latency human-robot interaction.	

Projects

Smart Retail Replenishment & Demand Forecasting System (SRR-DFS) |

Jan 2026 - Jan 2026

https://github.com/ankitghoshthecreator/Retail_Demand_Forecasting

Individual Project · Python, pandas, numpy, scikit-learn, matplotlib, seaborn, Streamlit, Time-series lag features, Moving Average, Safety Stock & Reorder Point modeling

Data Science

- Developed a SKU-level demand forecasting and inventory optimization system using Python and Random Forest, improving replenishment accuracy and reducing stockout risk.
- Implemented ABC inventory classification and automated reorder point calculations with safety stock modeling to enhance inventory turnover efficiency.
- Built an interactive Streamlit dashboard delivering real-time demand forecasts, sell-through analytics, and actionable restocking recommendations for retail operations.

Exactly-Once Payment Processing System with Distributed Ledger

Dec 2025 - Feb 2026

Consistency | <https://github.com/ankitghoshthecreator/transactionEngine>

Individual project · Python, Pytorch, OpenCV, Numpy, Pandas, MediaPipe, Scikit Learn

Java Backend Development

Designed a fault-tolerant financial transaction system ensuring exactly-once money movement across distributed services using idempotency, state machines, and ledger-based consistency.

- Designed a **distributed payment processing architecture** to guarantee exactly-once transaction execution under retries, failures, and concurrent requests
- Implemented a **transaction state machine** (INIT → AUTHORIZED → SETTLED → FAILED) to enforce deterministic financial state transitions
- Built an **idempotency layer** to prevent duplicate debits/credits during network retries and API replays

Document Intelligence Tool | [Link](#)

Apr 2026 - Present

· FastAPI · React · LangChain · FAISS · sentence-transformers · Gemini Flash · Python

Machine Learning, Web Development

- Built a production-ready RAG-based document intelligence system aligned with enterprise workflows similar to Zoho Docs, WorkDrive, and Sign — engineering the full pipeline from PDF ingestion to grounded Q&A using PyMuPDF, LangChain, sentence-transformers (all-MiniLM-L6-v2), and FAISS vector search.
- Implemented automated document summarization and structured key clause extraction (payment terms, obligations, deadlines) rendered as a JSON-backed table — directly applicable to contract intelligence use cases like Zoho Sign.
- Designed a context-aware chat interface with source attribution and chunk-level confidence scoring for answer grounding, deployed on Render (backend) and Vercel (frontend) with a modular, production-grade architecture.

UNITY- HandGestureControlledGame | [Link](#)

Feb 2025 - Present

· Python, MediaPipe, PyTorch, Unity, C#

Game

- Developed a two-player 2D Unity game integrating **hand gesture recognition** via **MediaPipe** and **PyTorch-based reinforcement learning** for adaptive gameplay.
- Designed and trained **RL agents** to learn gesture-response patterns, improving accuracy and player experience over time.
- Optimized real-time inference, gesture-to-action mapping, and gameplay performance for smooth and responsive multiplayer interaction.

Phaadu - A Math Heavy Programming Language | <https://github.com/ankitghoshthecreator/Phaadu>

Nov 2025 - Present

Individual · rust

Programming Language

1. **Built a custom programming language from first principles**, using a formally defined grammar and a structured tokenization system that handles user-defined symbols and operator hierarchies.
2. **Implemented a lexer–parser pipeline based on fundamental automata and grammar concepts**, converting raw text into a validated AST while maintaining predictable parsing behavior.
3. **Developed a deterministic interpreter using well-defined evaluation rules**, executing expressions, managing variable bindings, and ensuring consistent runtime semantics.
4. **Designed the architecture for extensibility**, allowing new syntax constructs, operators, and future compilation targets without rewriting core components.
5. **Added robust error diagnostics**, using structured position tracking and controlled failure states to generate clear syntax and runtime error messages.

Certifications

Programming in Java https://drive.google.com/file/d/1zhf_9a_gtrkqKoErrqI9Ibk773OAvJfw/view?usp=drivesdk NPTEL · NPTEL24CS105S252505015	Nov 2024
Data Base Management System https://drive.google.com/file/d/1skKhKAEoL0MXDhNYULJ65bPT-jH1eaUq/view?usp=drivesdk NPTEL · NPTEL25CS145S533211819	Sep 2025

Publications

Adaptive Portfolio Optimization using Robbins–Monro Stochastic Approximation: https://zenodo.org/records/17559546 zenodo	Nov 2025
- Developed an adaptive portfolio optimization framework using Robbins–Monro stochastic approximation, enabling real-time weight updates under non-stationary market conditions. - Implemented volatility-aware learning via EWMA-based modeling, dynamically tuning risk penalty (λ) and learning rate (η) to balance return maximization and risk control. - Evaluated model on 10+ years of financial data (Yahoo Finance), achieving robust performance (22.8% annualized return, Sharpe $\approx$ 1.0) with improved stability over static portfolio baselines.	

Skills

Programming Languages: Python, Java, GO, C/C++
Tools and Technologies: OpenCV, Transformers, YOLO, pytorch, Mediapupe, Streamlit
Areas of Interest: Natural Language Processing
Languages and Frameworks: Spring Boot
Design and Simulation Tools: Unity C#
Domain Knowledge: Speaker Identification

Languages

English [Professional Working Proficiency], Bengali [Native Proficiency], Hindi [Native Proficiency], Russian [Limited Working Proficiency], Chinese [Fundamental Proficiency], Korean [Fundamental Proficiency], Japanese [Elementary Proficiency]

Links

GitHub , LinkedIn , Instagram , LeetCode , HackerRank , Link
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